

5. Satellites are used to communicate between base stations at different places on Earth.

- (a) (i) **Complete the diagram** below to show how base station A communicates with base station B. [3]



- (ii) Explain why a geostationary satellite must be used for constant communication rather than a geosynchronous one. [2]

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- (b) A base station communicates with another base station using a single geostationary satellite 3.6×10^4 km above the Earth. They communicate with each other using microwaves of wavelength 2.8 cm travelling at 3×10^8 m/s.

Use an equation from page 2 to calculate the time delay between sending a signal from one base station and receiving the signal at the other. [4]

Time = s

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